Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam D · Form 2: Pulley Focus

Multiple Choice Questions: Select the best answer. (1 point each)

1. A sliding cart is pulled toward a pulley by a descending hanger. The same cart is then released from rest. Which parameter becomes necessary to decide whether motion starts?

(A) The old cart speed alone

(B) The old kinetic friction force alone

(C) The height of the pulley axle alone

(D) Maximum static friction

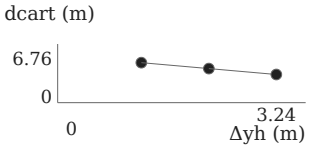
2. A cart and a hanging mass share a taut rope over one ideal fixed pulley. Measure downward hanger displacement and the cart distance from the pulley. Double the initial cart distance at the same hanger displacements. What is the new ratio of vertical change to the same horizontal change?

(A) -0.79

(B) -1.42

(C) -0.37

(D) -1



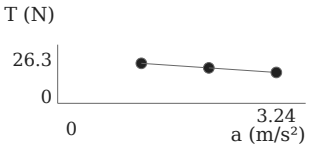
3. An unseen hanger rides on one side of an ideal pulley. Vary the opposite load and record tension with the unseen hanger acceleration positive downward. Use g=10 m/s². a (m/s²) -> T (N): (1.24, 26.28); (2.24, 23.28); (3.24, 20.28). Which comparison distinguishes vertical zero bias from slope error?

(A) Repeat at only one setting.

(B) Force every fit through the origin.

(C) Compare observed minus predicted residuals at separated settings.

(D) Remove units before comparing.



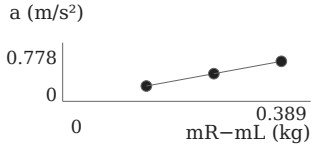
4. Transfer mass between two sides of an ideal Atwood pulley, keeping total mass 5 kg. Right downward is positive; g=10 m/s². Add equal masses to both pans to double total mass, testing the same imbalances. Under the revised model, what is Y at X=0.2688?

(A) 1.529

(B) 0.2688

(C) 0.6888

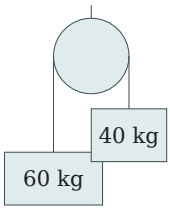
(D) -0.5712



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A motorless 60 kg cabin and a 40 kg counterweight share an ideal pulley. The cabin is moving upward; use g=10 m/s².

a. Mark the cabin velocity and net-force directions. **(1 pt)**



b. Find the acceleration and cable tension and explain why tension differs from cabin weight. **(3 pt)**

6. A horizontal cart slides toward an ideal pulley. Cart plus hanger mass is fixed at 5 kg; a constant 1 N resistance acts on the cart. Use g=10 m/s². Wh (N) -> a (m/s²): (1.24, 0.048); (2.24, 0.248); (3.24, 0.448).

a. Build the model for sliding cart and hanger from physics and predict both slope and intercept. **(2 pt)**

b. Double the resistance at fixed system mass. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. An incline block of mass M slides downhill, lifting a hanging mass m. The incline angle is θ and kinetic coefficient is μ. Derive acceleration with downhill positive for the incline block. **(3 pt)**